

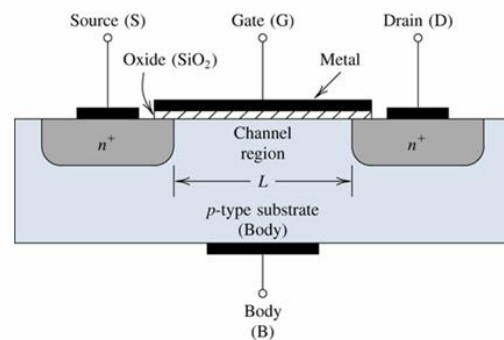
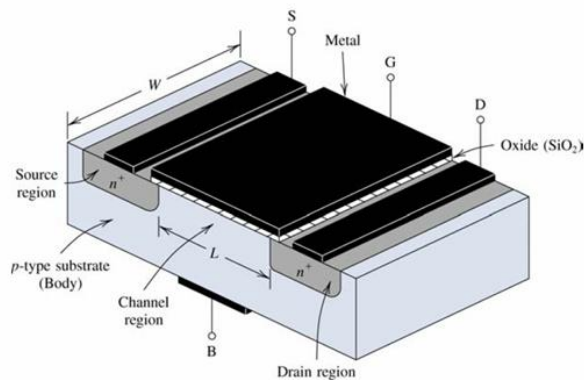
Lab 2

Name: Shiva Kumar

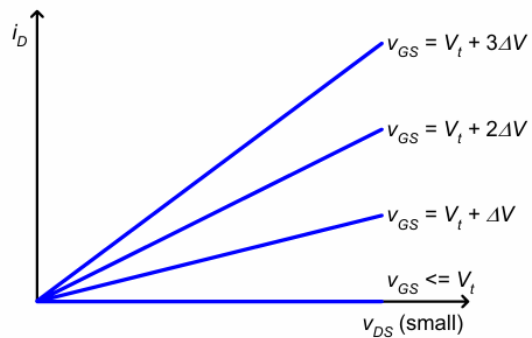
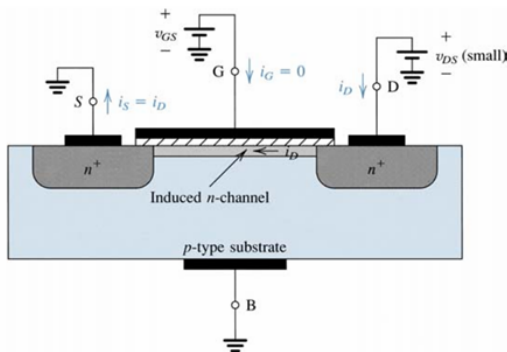
Enrollment No: 92301733056

Experiment: Hands-on experimentation on the current-voltage characteristics of NMOS transistor and PMOS transistor and find power dissipation

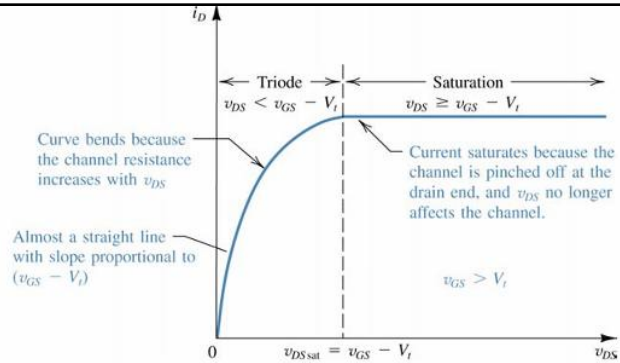
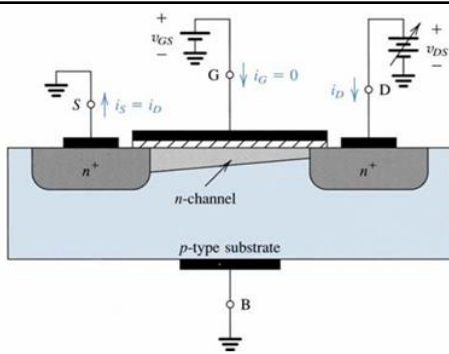
Device Structure and Physical Operation



Linear Operation



Triode to Saturation Region



Cutoff Mode:

Occurs when $V_{GS} \leq V_{TH(N)}$

$$I_D = 0$$

A MOSFET is in cutoff when no channel has been induced. Thus, for an enhancement NMOS device:

$$\text{if } v_{GS} - V_t < 0 \text{ then NMOS in CUTOFF}$$

Likewise, for an enhancement PMOS device:

$$\text{if } v_{GS} - V_t > 0 \text{ then PMOS in CUTOFF}$$

Triode Mode:

Occurs when $V_{GS} > V_{TH(N)}$ and $V_{DS} < V_{GS} - V_{TH(N)}$

$$I_D = \frac{W}{L} \mu_n C_{OX} \left(V_{GS} - V_{TH(N)} - \frac{V_{DS}}{2} \right) V_{DS}$$

$$\text{if } v_{GS} - V_t > 0 \text{ and } v_{DS} < v_{GS} - V_t \text{ then NMOS in TRIODE}$$

$$\text{if } v_{GS} - V_t < 0 \text{ and } v_{DS} > v_{GS} - V_t \text{ then PMOS in TRIODE}$$

Saturation Mode:

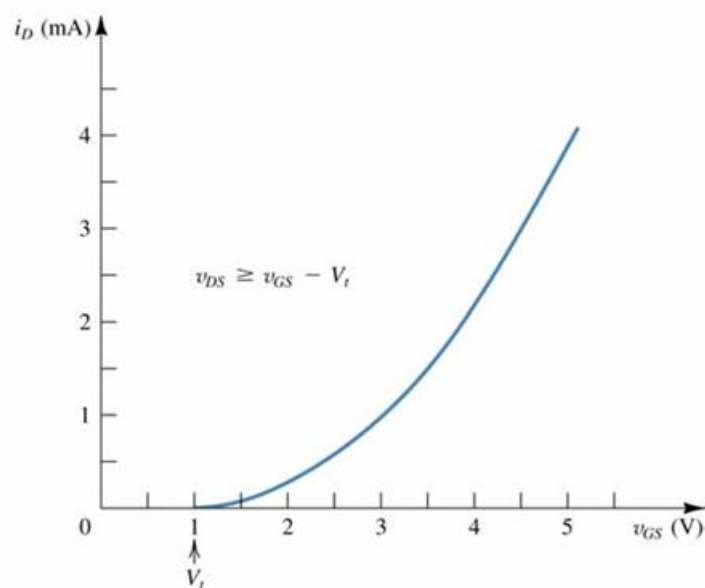
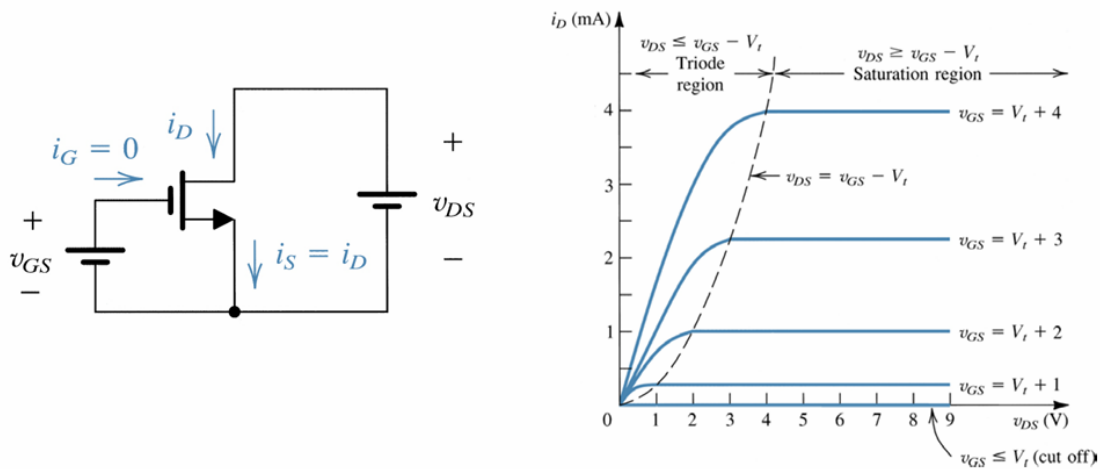
Occurs when $V_{GS} > V_{TH(N)}$ and $V_{DS} \geq V_{GS} - V_{TH(N)}$

$$I_D = \frac{W}{L} \mu_n C_{OX} \frac{1}{2} (V_{GS} - V_{TH(N)})^2 (1 + \lambda_n V_{DS})$$

if $v_{GS} - V_t > 0$ and $v_{DS} > v_{GS} - V_t$ then NMOS in SAT.

if $v_{GS} - V_t < 0$ and $v_{DS} < v_{GS} - V_t$ then PMOS in SAT.

Circuit



The Process Transconductance Parameter k' is a constant that depends on the process technology used to fabricate an integrated circuit. Therefore, all the transistors on a given substrate will typically have the same value of this parameter.

The Channel Aspect Ratio WL is simply the ratio of channel width W to channel length L . This is the MOSFET device parameter that can be altered and modified by the circuit designer to satisfy the requirements of the given circuit or transistor.

We can likewise combine these parameter to form a single MOSFET device parameter K :

$$K = \frac{1}{2} k' \left(\frac{W}{L} \right) \quad \left[\frac{A}{V^2} \right]$$

Observation Table:

NMOS

Model Parameters:

$$V_T = 0.7V \quad W = 10\mu m \quad L = 1\mu m \quad W/L = 10$$

$$V_{DS} = 0.2$$

V_{GS}	I_D
0V	0mA
0.7V	0mA
0.8V	0.006mA
1.2V	0.050mA
2.2V	0.299mA
3V	0.528mA

$$V_{DS} = 2.5$$

V_{GS}	I_D
0V	0mA
0.7V	0mA
0.8V	0.06mA
1.2V	0.059mA
2.2V	1.023mA
3.3V	4.050mA

$$V_{DS} = 5$$

V_{GS}	I_D
0V	0mA
0.7V	0mA
0.8V	0.006mA
2.2V	1.019mA
4V	6.554mA
5.8V	15.600mA

PMOS

Model Parameters:

$$V_T = -0.7V \quad W = 10\mu m \quad L = 1\mu m \quad W/L=10$$

$$V_{DS} = 0.2$$

$$V_{DS} = 2.5$$

$$V_{DS} = 5$$

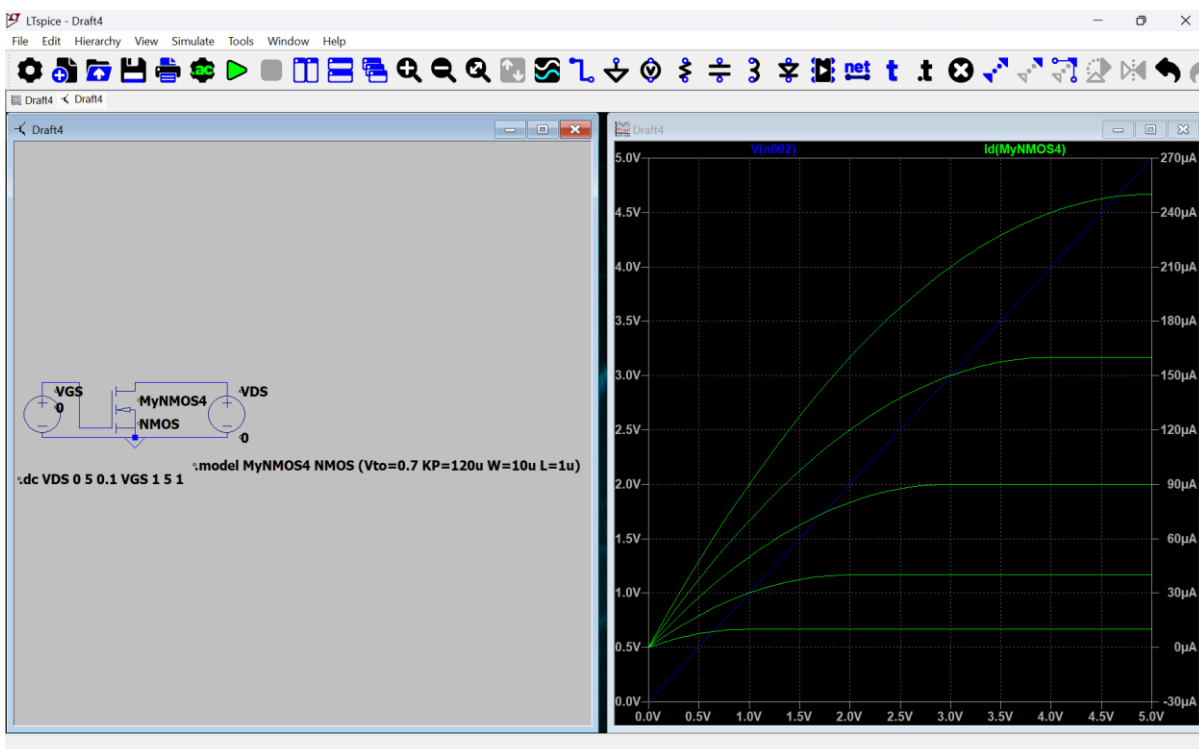
V_{GS}	I_D
0V	0mA
-0.7V	0mA
-0.9V	-0.004mA
-1V	-0.024mA
-2.2V	-0.150mA
-3V	-0.264 mA

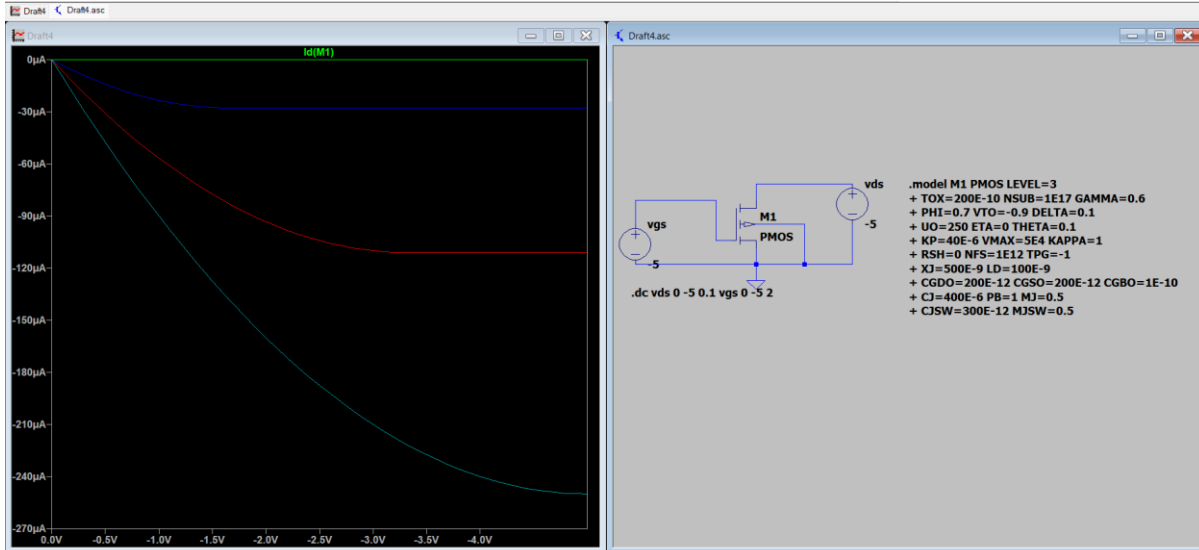
V_{GS}	I_D
0V	0mA
-0.7V	0mA
-0.8V	-0.004mA
-1V	-0.027mA
-2.2V	-0.518mA
-3.3V	-2.025mA

V_{GS}	I_D
0V	0mA
-0.7V	0mA
-0.8V	-0.004mA
-2.2V	-0.515mA
-5.2V	-5.599mA
-5.8V	-7.800mA

Output Waveform

NMOS:



PMOS:**Conclusion:**

I conclude that I performed the simulation of Hands-on experimentation on the current-voltage characteristics of NMOS transistor and PMOS transistor and find power dissipation